D3.1 Reproduction (HL)

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 NAME the hormone from the hypothalamus whose rise triggers puberty. [D3.1.13] [1]

2 STATE how many sperm develop from one primary spermatocyte. [D3.1.14] [1]

3 NAME the glycoprotein coat that surrounds the egg. [D3.1.15] [1]

4 NAME the hormone detected by a pregnancy test. [D3.1.17] [1]

5 NAME the hollow ball of cells that implants in the endometrium. [D3.1.16] [1]

6 STATE two substances that pass from the mother to the foetus across the placenta. [D3.1.18] [2]

7 NAME the hormone that stimulates contractions of the uterus in childbirth. [D3.1.19] [1]

Section B: Understanding

15 MARKS

8 COMPARE spermatogenesis and oogenesis. [D3.1.14] [4]

9 EXPLAIN how polyspermy is prevented. [D3.1.15] [3]

10 **EXPLAIN** how a pregnancy test shows that a woman is pregnant. [D3.1.17] [3]

11 **EXPLAIN** the role of progesterone during pregnancy. [D3.1.19] [2]

12 **EXPLAIN** why the control of childbirth is an example of positive feedback. [D3.1.19] [3]

Section C: Working with evidence

11 MARKS

All data in this section are invented for practice. The pattern matches real findings.

13 [D3.1.20]

Two studies looked at hormone replacement therapy (HRT) and coronary heart disease (CHD) in women aged 50 to 70.

STUDY	GROUP	CHD CASES PER 1000 WOMEN PER YEAR	WOMEN WITH A UNIVERSITY DEGREE / %
A: observational (women chose)	Took HRT	3.0	62
	No HRT	4.5	35
B: randomized trial, 8000 per group	Given HRT	3.9	48
	Given placebo	3.4	48

a CALCULATE the percentage by which the CHD rate was lower in HRT users in study A. [2]

b COMPARE the conclusions that could be drawn from the two studies. [2]

c EXPLAIN, using the data, why the result of study A may be misleading. [3]

d EXPLAIN why random allocation gives more reliable evidence. [2]

e SUGGEST why study B needed thousands of women in each group. [2]

Section D: Longer answer

6 MARKS

14 DESCRIBE oogenesis, from the ovary of a female foetus to fertilization. [D3.1.14] [6]